

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VII

Roll No. : 42	Name: Aniket Tiwari
Class : SYIT	Batch : 00
Date of Assignment : 05-08-2026	Date/Time of Submission :05-08-2026

Hash Table:

Write a program to:

- a. Implement a hash table with separate chaining for collision handling.

CODE:

```
size = 5
table = [[] for _ in range(size)]

def insert(key,value):
    table[(key%size)].append((key,value))

insert(10,"Apple")
insert(15,"Banana")
insert(20,"Mango")
insert(32,"Guava")
insert(33,"Pineapple")

print(table)
print("Aniket")
```

OUTPUT:

```
===== RESTART: D:/Aniket SYIT/
DS Prac 7.py =====
[(10, 'Apple'), (15, 'Banana'), (20, 'Mango')
], [], [(32, 'Guava')], [(33, 'Pineapple')], [
]]
Aniket
```

- b. Store and retrieve data from the hash table.

CODE:

```
size = 5
table = [[] for _ in range(size)]

def insert(key,value):
    table[(key%size)].append((key,value))

def search(key):
    for k,v in table[key%size]:
        if k==key:
            return v
    return "Not Found"

insert(10,"Apple")
insert(15,"Banana")
insert(20,"Mango")
insert(32,"Guava")
insert(33,"Pineapple")

print(table)

key = int(input("Enter Key : "))
print(search(key))

print("Aniket")
```

OUTPUT:

```
==== RESTART: D:/Aniket SYIT/DS Prac 7.py ====
[[ (10, 'Apple'), (15, 'Banana'), (20, 'Mango')
], [], [(32, 'Guava')], [(33, 'Pineapple')], [
]]
Enter Key : 12
Not Found
Aniket
```

```
==== RESTART: D:/Aniket SYIT/DS Prac 7.py ====
[[ (10, 'Apple'), (15, 'Banana'), (20, 'Mango')
], [], [(32, 'Guava')], [(33, 'Pineapple')], [
]]
Enter Key : 10
Apple
Aniket
```